Password Cracking Simulator – Complete Code Explanation

This document explains the complete structure and working of the Password Cracking Simulator GUI application written in Python using Tkinter.

Lines 1–24: Imports and Documentation

- Shebang allows running the script directly on Linux/Mac.

- Docstring describes the tool purpose.

- hashlib for hashing (MD5, SHA-1, SHA-256).

- itertools and string for brute-force combinations.

- time for measuring attack speed.

- typing for clean type hints.

- tkinter and ttk for GUI.

- threading to avoid GUI freezing.

PasswordCracker Class Initialization

- Tracks attempts, start time, rainbow table, and stop flag.

Hash Password Method

- Converts plain password into MD5, SHA-1, or SHA-256 hash.

Brute Force Attack

- Tries all combinations up to selected length.

- Updates GUI with progress.

- Stops safely when requested.

Dictionary Attack

- Uses common passwords and variations.

- Efficient for weak passwords.

Rainbow Table Attack

- Uses precomputed hash-password pairs.

- Instant lookup if present in table.

Hybrid Attack

- Dictionary words combined with numbers and special characters.

Password Strength Analyzer

- Scores password based on length, uppercase, lowercase, digits, and symbols.
- Provides recommendations.

Helper Methods

- Generate rainbow table.
- Provide sample wordlist.
- Create variations of words.

GUI (PasswordCrackerGUI)

- Modern dark theme UI.
- Left panel for configuration.
- Right panel for results.
- Buttons for starting, stopping attacks, and analyzing passwords.

Generate Hash

- Converts entered password to SHA-256 and displays it.

Start Attack

- Runs selected attack in background thread.

Run Attack

- Calls appropriate attack method.

Stop and Completion Handling

- Safely stops attack and restores UI state.

Password Analysis Display

- Shows strength, characteristics, and recommendations.

Logging and Clear Functions

- Handles colored output and clearing results.

Main Function

- Initializes Tkinter app and starts main loop.